Bases de Données Fall, 2026

TP 4 (universityDB-data)

1 Setting up PostgreSQL (if not done already)

USING USPN LABS DATABASE

- Login to `linux` with your user account.
- Type the following command to start the `postgreSQL prompt`:

  ```
  psql -h postgres -U m1info_xx
  ```

  where `xx` is your account number between 1 and 90. The password is `M1INFO_xx`, with the same value of `xx` as your account number.
- You can see the list of tables present by typing :

  ```
  \d
  ```

2 Exercises

Now write the `SQL` code, and then run to verify, the following queries. The table after each question gives the result that you should get.

1. For all teachers, give their `ID`, `name`, `dept_name`, `salary`, and a sequential number of that teacher (in order of decreasing salary *inside their own department*). Use **window functions**.

id	*name*	*dept_name*	*salary*	*num*
76766	Crick	Biology	72000.00	1
83821	Brandt	Comp. Sci.	92000.00	1
45565	Katz	Comp. Sci.	75000.00	2
10101	Srinivasan	Comp. Sci.	65000.00	3
98345	Kim	Elec. Eng.	80000.00	1
12121	Wu	Finance	90000.00	1
76543	Singh	Finance	80000.00	2
58583	Califieri	History	62000.00	1
32343	El Said	History	60000.00	2
15151	Mozart	Music	40000.00	1
22222	Einstein	Physics	95000.00	1
33456	Gold	Physics	87000.00	2

2. For each department, give the number of courses it offers, as well as the rank of that number among all departments. Use **window functions**.

   Just the first 10 lines of the solution table:

dept_name	*count*	*rank*
Comp. Sci.	5	1
Biology	3	2
Elec. Eng.	1	3
Finance	1	3
History	1	3
Music	1	3
Physics	1	3

3. For each department, give its `dept_name`, `budget`, and the percentage that its budget represents of the total budget of all departments. Use **window functions**.

   Just the first 10 lines of the solution table:

dept_name	*budget*	*pct*
Biology	90000	15.126
Comp. Sci.	100000	16.807
Elec. Eng.	85000	14.286
Finance	120000	20.168
History	50000	8.403
Music	80000	13.445
Physics	70000	11.765

4. For each teacher, give their `name`, `dept_name`, `salary`, the salary of the best-paid teacher in their department, and the salary of the worst-paid teacher in their department. Use **window functions**.

   Just the first 10 lines of the solution table:

name	*dept_name*	*salary*	*highest*	*lowest*
Crick	Biology	72000.00	72000.00	72000.00
Brandt	Comp. Sci.	92000.00	92000.00	65000.00
Katz	Comp. Sci.	75000.00	92000.00	65000.00
Srinivasan	Comp. Sci.	65000.00	92000.00	65000.00
Kim	Elec. Eng.	80000.00	80000.00	80000.00
Wu	Finance	90000.00	90000.00	80000.00
Singh	Finance	80000.00	90000.00	80000.00
Califieri	History	62000.00	62000.00	60000.00
El Said	History	60000.00	62000.00	60000.00
Mozart	Music	40000.00	40000.00	40000.00
Einstein	Physics	95000.00	95000.00	87000.00
Gold	Physics	87000.00	95000.00	87000.00

5. For each teacher, give their `name`, `salary`, and the quartile (1 to 4) into which their salary falls when all salaries are ordered from highest to lowest. Use **window functions**.

   Just the first 10 lines of the solution table:

name	*salary*	*quartile*
Einstein	95000.00	1
Brandt	92000.00	1
Wu	90000.00	1
Gold	87000.00	2
Singh	80000.00	2
Kim	80000.00	2
Katz	75000.00	3
Crick	72000.00	3
Srinivasan	65000.00	3
Califieri	62000.00	4
El Said	60000.00	4
Mozart	40000.00	4

6. For each student who has taken at least one course, give their `ID`, `name`, the number of courses they have taken, and the rank of that number among all students. Use **window functions**.

   Just the first 10 lines of the solution table:

ID	*name*	*num_courses*	*rank*
12345	Shankar	4	1
45678	Levy	3	2
00128	Zhang	2	3
54321	Williams	2	3
76543	Brown	2	3
98765	Bourikas	2	3
98988	Tanaka	2	3
19991	Brandt	1	8
23121	Chavez	1	8
44553	Peltier	1	8

7. Give the `name`, `dept_name` and `salary` of the two best-paid teachers in each department. Use **window functions**.

   Just the first 10 lines of the solution table:

name	*dept_name*	*salary*
Crick	Biology	72000
Brandt	Comp. Sci.	92000
Katz	Comp. Sci.	75000
Kim	Elec. Eng.	80000
Wu	Finance	90000
Singh	Finance	80000
Califieri	History	62000
El Said	History	60000
Mozart	Music	40000
Einstein	Physics	95000

8. For each student, give their `name`, `dept_name`, `tot_cred`, the average total credits of their department, the difference between their credits and that average, and the rank of their credits inside their department. Use **window functions**.

   Just the first 10 lines of the solution table:

name	*dept_name*	*tot_cred*	*dept_avg*	*diff*	*rnk*
Tanaka	Biology	120	120	0	1
Zhang	Comp. Sci.	102	61.5	40.5	1
Brown	Comp. Sci.	58	61.5	-3.5	2
Williams	Comp. Sci.	54	61.5	-7.5	3
Shankar	Comp. Sci.	32	61.5	-29.5	4
Bourikas	Elec. Eng.	98	79	19	1
Aoi	Elec. Eng.	60	79	-19	2
Chavez	Finance	110	110	0	1
Brandt	History	80	80	0	1
Sanchez	Music	38	38	0	1

9. For each student who has at least one graded course, compute their GPA (weighted average of grade points by credits, using the scale A=4.0, A-=3.7, B+=3.3, B=3.0, B-=2.7, C+=2.3, C=2.0, C-=1.7, F=0.0). Then, for each such student, give their `ID`, `name`, `dept_name`, their GPA, the average GPA of their department, and the rank of their GPA inside their department. Use **window functions**.

   Just the first 10 lines of the solution table:

ID	*name*	*dept_name*	*gpa*	*dept_avg_gpa*	*rnk*
98988	Tanaka	Biology	4.000	4.000	1
76543	Brown	Comp. Sci.	4.000	3.700	1
00128	Zhang	Comp. Sci.	3.871	3.700	2
54321	Williams	Comp. Sci.	3.500	3.700	3
12345	Shankar	Comp. Sci.	3.429	3.700	4
98765	Bourikas	Elec. Eng.	2.257	2.129	1
76653	Aoi	Elec. Eng.	2.000	2.129	2
23121	Chavez	Finance	2.300	2.300	1
19991	Brandt	History	3.000	3.000	1
55739	Sanchez	Music	3.700	3.700	1

10. For each section, give its building, `course_id`, `sec_id`, `semester`, `year`, the capacity of its classroom, the number of enrolled students, the utilization percentage of the classroom, and the rank of the enrollment among the sections in the same building. Use **window functions**.

    Just the first 10 lines of the solution table:

building	*course_id*	*sec_id*	*semester*	*year*	*capacity*	*enroll*	*util_pct*	*bld_rank*
Packard	CS-101	1	Fall	2009	500	6	1.200	1
Packard	CS-101	1	Spring	2010	500	1	0.200	2
Packard	FIN-201	1	Spring	2010	500	1	0.200	2
Packard	MU-199	1	Spring	2010	500	1	0.200	2
Painter	BIO-101	1	Summer	2009	10	1	10.000	1
Painter	BIO-301	1	Summer	2010	10	1	10.000	1
Painter	HIS-351	1	Spring	2010	10	1	10.000	1
Taylor	CS-190	2	Spring	2009	70	2	2.857	1
Taylor	CS-347	1	Fall	2009	70	2	2.857	1
Taylor	CS-319	2	Spring	2010	70	1	1.429	3

11. For each advisor (a teacher who advises at least one student), give their `ID`, `name`, `salary`, the number of advisees, the average total credits of their advisees, the rank of their salary among all teachers, and the rank of their advisee count among all advisors. Use **window functions**.

    Just the first 10 lines of the solution table:

i_ID	*name*	*salary*	*advisees*	*avg_cred*	*salary_rank*	*load_rank*
22222	Einstein	95000	2	51	1	1
98345	Kim	80000	2	79	5	1
45565	Katz	75000	2	80	7	1
76543	Singh	80000	1	110	5	4
76766	Crick	72000	1	120	8	4
10101	Srinivasan	65000	1	32	9	4

12. Give the `name`, `dept_name` and `tot_cred` of the top-2 ranked students (by total credits) in each department, including ties in the ranks. Use **window functions**.

    Just the first 10 lines of the solution table:

name	*dept_name*	*tot_cred*	*rnk*
Tanaka	Biology	120	1
Zhang	Comp. Sci.	102	1
Brown	Comp. Sci.	58	2
Bourikas	Elec. Eng.	98	1
Aoi	Elec. Eng.	60	2
Chavez	Finance	110	1
Brandt	History	80	1
Sanchez	Music	38	1
Peltier	Physics	56	1
Levy	Physics	46	2

13. For each department, give its `dept_name`, `building`, `budget`, the rank of its budget among the departments in the same building, and the largest budget in that building. Use **window functions**.

    Just the first 10 lines of the solution table:

dept_name	*building*	*budget*	*rnk*	*bld_max*
Music	Packard	80000	1	80000
Finance	Painter	120000	1	120000
History	Painter	50000	2	120000
Comp. Sci.	Taylor	100000	1	100000
Elec. Eng.	Taylor	85000	2	100000
Biology	Watson	90000	1	90000
Physics	Watson	70000	2	90000

14. For each graded enrollment, give the student's `ID`, the `course_id`, `semester`, `year`, `grade`, and the student's *running GPA* at that point in time: the weighted average of all grade points earned so far, in chronological order (Spring, Summer, Fall, Winter). Use **window functions**.

    Just the first 10 lines of the solution table:

ID	*course_id*	*semester*	*year*	*grade*	*running_gpa*
00128	CS-101	Fall	2009	A	4.000
00128	CS-347	Fall	2009	A-	3.871
12345	CS-190	Spring	2009	A	4.000
12345	CS-101	Fall	2009	C	3.000
12345	CS-347	Fall	2009	A	3.273
12345	CS-315	Spring	2010	A	3.429
19991	HIS-351	Spring	2010	B	3.000
23121	FIN-201	Spring	2010	C+	2.300
44553	PHY-101	Fall	2009	B-	2.700
45678	CS-101	Fall	2009	F	0.000

Tables

course_id	*title*	*dept_name*	*credits*
BIO-101	Intro. to Biology	Biology	4
BIO-301	Genetics	Biology	4
BIO-399	Computational Biology	Biology	3
CS-101	Intro. to Computer Science	Comp. Sci.	4
CS-190	Game Design	Comp. Sci.	4
CS-315	Robotics	Comp. Sci.	3
CS-319	Image Processing	Comp. Sci.	3
CS-347	Database System Concepts	Comp. Sci.	3
EE-181	Intro. to Digital Systems	Elec. Eng.	3
FIN-201	Investment Banking	Finance	3
HIS-351	World History	History	3
MU-199	Music Video Production	Music	3
PHY-101	Physical Principles	Physics	4

(a) course

id	*name*	*dept_name*	*salary*
10101	Srinivasan	Comp. Sci.	65000.00
12121	Wu	Finance	90000.00
15151	Mozart	Music	40000.00
22222	Einstein	Physics	95000.00
32343	El Said	History	60000.00
33456	Gold	Physics	87000.00
45565	Katz	Comp. Sci.	75000.00
58583	Califieri	History	62000.00
76543	Singh	Finance	80000.00
76766	Crick	Biology	72000.00
83821	Brandt	Comp. Sci.	92000.00
98345	Kim	Elec. Eng.	80000.00

(b) teacher

id	*name*	*dept_name*	*tot_cred*
00128	Zhang	Comp. Sci.	102
12345	Shankar	Comp. Sci.	32
19991	Brandt	History	80
23121	Chavez	Finance	110
44553	Peltier	Physics	56
45678	Levy	Physics	46
54321	Williams	Comp. Sci.	54
55739	Sanchez	Music	38
70557	Snow	Physics	0
76543	Brown	Comp. Sci.	58
76653	Aoi	Elec. Eng.	60
98765	Bourikas	Elec. Eng.	98
98988	Tanaka	Biology	120

(c) student

course_id	*sec_id*	*semester*	*year*	*building*	*rn*	*time_id*
BIO-101	1	Summer	2009	Painter	514	B
BIO-301	1	Summer	2010	Painter	514	A
CS-101	1	Fall	2009	Packard	101	H
CS-101	1	Spring	2010	Packard	101	F
CS-190	1	Spring	2009	Taylor	3128	E
CS-190	2	Spring	2009	Taylor	3128	A
CS-315	1	Spring	2010	Watson	120	D
CS-319	1	Spring	2010	Watson	100	B
CS-319	2	Spring	2010	Taylor	3128	C
CS-347	1	Fall	2009	Taylor	3128	A
EE-181	1	Spring	2009	Taylor	3128	C
FIN-201	1	Spring	2010	Packard	101	B
HIS-351	1	Spring	2010	Painter	514	C
MU-199	1	Spring	2010	Packard	101	D
PHY-101	1	Fall	2009	Watson	100	A

(d) section

id	*course_id*	*sec_id*	*semester*	*year*
10101	CS-101	1	Fall	2009
10101	CS-315	1	Spring	2010
10101	CS-347	1	Fall	2009
12121	FIN-201	1	Spring	2010
15151	MU-199	1	Spring	2010
22222	PHY-101	1	Fall	2009
32343	HIS-351	1	Spring	2010
45565	CS-101	1	Spring	2010
45565	CS-319	1	Spring	2010
76766	BIO-101	1	Summer	2009
76766	BIO-301	1	Summer	2010
83821	CS-190	1	Spring	2009
83821	CS-190	2	Spring	2009
83821	CS-319	2	Spring	2010
98345	EE-181	1	Spring	2009

(e) teaches

id	*course_id*	*sec_id*	*semester*	*year*	*grade*
00128	CS-101	1	Fall	2009	A
00128	CS-347	1	Fall	2009	A-
12345	CS-101	1	Fall	2009	C
12345	CS-190	2	Spring	2009	A
12345	CS-315	1	Spring	2010	A
12345	CS-347	1	Fall	2009	A
19991	HIS-351	1	Spring	2010	B
23121	FIN-201	1	Spring	2010	C+
44553	PHY-101	1	Fall	2009	B-
45678	CS-101	1	Fall	2009	F
45678	CS-101	1	Spring	2010	B+
45678	CS-319	1	Spring	2010	B
54321	CS-101	1	Fall	2009	A-
54321	CS-190	2	Spring	2009	B+
55739	MU-199	1	Spring	2010	A-
76543	CS-101	1	Fall	2009	A
76543	CS-319	2	Spring	2010	A
76653	EE-181	1	Spring	2009	C
98765	CS-101	1	Fall	2009	C-
98765	CS-315	1	Spring	2010	B
98988	BIO-101	1	Summer	2009	A
98988	BIO-301	1	Summer	2010	

(f) takes

dept_name	*building*	*budget*
Biology	Watson	90000.00
Comp. Sci.	Taylor	100000.00
Elec. Eng.	Taylor	85000.00
Finance	Painter	120000.00
History	Painter	50000.00
Music	Packard	80000.00
Physics	Watson	70000.00

(g) department

building	*room_number*	*capacity*
Packard	101	500
Painter	514	10
Taylor	3128	70
Watson	100	30
Watson	120	50

(h) classroom